



# EP

## Engineering & Planning

Read the game, define the work, plan the season.

7 cards EP-1 – EP-7

### WHAT BELONGS IN THIS FOLDER

|      |                                      |    |
|------|--------------------------------------|----|
| EP-1 | Team Identity: Name, Look & Motto    | •  |
| EP-2 | Read the Game, Build the Wall Poster | •  |
| EP-3 | Robot Requirements List              | •• |
| EP-4 | Design Sketches & Pick a Concept     | •• |
| EP-5 | Set Up the Engineering Notebook      | •  |
| EP-6 | Tape Out the Practice Field          | •  |
| EP-7 | Shop Safety & Tool Check-Off         | •  |

**Taking a sheet?** Move your chip to CLAIMED on the board and write your names next to it.

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**Ran out of time?** Note on the back where you stopped and put it in **IN PROGRESS**.



# CD

## CAD & 3D Printing

Model it in Onshape, then make it real.

**8 cards** CD-1 – CD-8

### WHAT BELONGS IN THIS FOLDER

|      |                                                  |     |
|------|--------------------------------------------------|-----|
| CD-1 | Onshape Setup & Your First Part                  | •   |
| CD-2 | Measure & Model a Real Robot Part                | ••  |
| CD-3 | Design a Pollen Intake in CAD                    | ••• |
| CD-4 | Design a Pollen Outtake in CAD                   | ••• |
| CD-5 | Build the Robot in CAD: Assembly & Fit Check     | ••• |
| CD-6 | From CAD to Printed Part                         | ••  |
| CD-7 | Design a Printed Part That Solves a Real Problem | ••• |
| CD-8 | Design the Odometry Mounting Adapter             | ••• |

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# PR

## Programming

Make the robot do what you tell it.

17 cards PR-1 – PR-17

### WHAT BELONGS IN THIS FOLDER

|       |                                                |     |               |
|-------|------------------------------------------------|-----|---------------|
| PR-1  | Get Connected: Control Hub & Driver Station    | •   | BOARD / ROBOT |
| PR-2  | Build the Robot Configuration File             | •   | BOARD / ROBOT |
| PR-3  | First TeleOp: Make It Drive (Mecanum)          | •   | ROBOT         |
| PR-4  | Telemetry: See What the Robot Is Thinking      | ••  | BOARD / ROBOT |
| PR-5  | Driver Tuning: Slow Mode & Smooth Control      | ••  | ROBOT + FIELD |
| PR-6  | First Autonomous: Time-Based Movement          | ••  | ROBOT + FIELD |
| PR-7  | Encoder Autonomous: Drive an Exact Distance    | ••• | ROBOT + FIELD |
| PR-8  | Intake Control: Map a Mechanism to a Button    | ••  | ROBOT         |
| PR-9  | Practice Without the Robot: FTC SIM            | •   |               |
| PR-10 | Touch Sensor: The First Decision               | •   | BOARD         |
| PR-11 | Color Sensor: Teach It to See Pollen           | ••  | BOARD         |
| PR-12 | Potentiometer & Servo: Analog In, Position Out | ••  | BOARD         |
| PR-13 | Bench to Robot: Move Your Code Over            | ••• | ROBOT         |
| PR-14 | Field-Centric Drive with the IMU               | ••• | ROBOT + FIELD |
| PR-15 | Read the Pinpoint: Configure & Prove It        | ••  | ROBOT         |
| PR-16 | Pedro Pathing: Install & Tune the Localizer    | ••• | ROBOT + FIELD |
| PR-17 | Pedro Pathing: Tune the Follower               | ••• | ROBOT + FIELD |

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# BA

## Building & Assembly

Cut, bolt, wire, and make it hold together.

10 cards BA-1 – BA-10

### WHAT BELONGS IN THIS FOLDER

|       |                                               |     |       |
|-------|-----------------------------------------------|-----|-------|
| BA-1  | Inventory the Kit & Set Up the Shop           | •   |       |
| BA-2  | Mecanum Chassis: Build the Frame              | •   | ROBOT |
| BA-3  | Mecanum Drive: Gearboxes, Motors & Wheels     | ••  | ROBOT |
| BA-4  | Electronics: Mount & Wire the Control Hub     | ••  | ROBOT |
| BA-5  | Design & Build an Intake                      | ••• | ROBOT |
| BA-6  | Battery Management Station                    | •   |       |
| BA-7  | Scoring Mechanism Prototype (cardboard first) | ••• | ROBOT |
| BA-8  | Robot Cart & Pit Kit                          | •   | ROBOT |
| BA-9  | Assemble a Programming Board                  | •   | BOARD |
| BA-10 | Install the Pinpoint Odometry Pods            | ••  | ROBOT |

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# MG

## Merch & Give-Away

Things we hand people so they remember us.

**6 cards** MG-1 — MG-6

### WHAT BELONGS IN THIS FOLDER

|      |                                        |     |
|------|----------------------------------------|-----|
| MG-1 | Give-Away Plan & Budget                | •   |
| MG-2 | Design the Give-Away Art               | ••  |
| MG-3 | Produce the Give-Aways (assembly line) | •   |
| MG-4 | Team Card / Business Card              | ••  |
| MG-5 | 3D-Printed Give-Away                   | ••• |
| MG-6 | Design the Award Patch Set             | ••  |

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# TS

## Team T-Shirts

What the team looks like at an event.

4 cards TS-1 – TS-4

### WHAT BELONGS IN THIS FOLDER

|      |                                  |    |
|------|----------------------------------|----|
| TS-1 | Shirt Design Concepts            | •  |
| TS-2 | Vote & Make the Art Print-Ready  | •• |
| TS-3 | Sizing Survey & Order Sheet      | •  |
| TS-4 | Get Quotes & Recommend a Printer | •• |

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# MF

## Marketing & Fundraising

Find the money and tell the story.

7 cards MF-1 – MF-7

### WHAT BELONGS IN THIS FOLDER

|      |                                             |     |       |
|------|---------------------------------------------|-----|-------|
| MF-1 | Build the Sponsor Prospect List             | •   |       |
| MF-2 | Sponsorship One-Pager                       | ••  |       |
| MF-3 | Write the Outreach Emails & Letters         | ••  |       |
| MF-4 | Social Media Plan & First Posts             | •   |       |
| MF-5 | Team Page Content for the Herobotix Website | ••  |       |
| MF-6 | Plan a Demo / Outreach Event                | ••  | ROBOT |
| MF-7 | Grants & Team Funding Applications          | ••• |       |

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# IN PROGRESS

Started but not finished.

A sheet lives here when the meeting ended before the work did. Write on the back where you stopped and what the next crew needs to know. These get picked up before anyone starts something new.



# DONE & STAMPED

Every “Done when” box is true.

Stamped sheets wait here until they are logged and taped into the engineering notebook. A stamped sheet is already a notebook entry — it carries the checklist, the names and the date.